Practical Codes/NWT/5. Ethernet/EC.md

Experiment: Ethernet Cable Crimping (Straight and Cross Cable)

Theory:

1. **Ethernet Crimping** is the process of attaching **RJ-45 connectors** to both ends of a **UTP (Unshielded Twisted Pair)** cable so that it can be used for data transmission between network devices.

2. Ethernet cables are based on **Category 5e / Category 6** (Cat5e/Cat6) twisted-pair cables, containing **4 pairs (8 wires)** with specific color codes.

3. The two most common wiring standards are:

   - **T568A**
   - **T568B** These define the order of wire colors for proper connectivity.

4. The color sequence must be **consistent** to ensure proper transmission (Tx) and reception (Rx) signals.

5. **Straight-through Cable**:

   - Both ends use the **same wiring standard** (T568A–T568A or T568B–T568B).
   - Used to connect **different devices** (e.g., PC ↔ Switch, Switch ↔ Router).

6. **Cross-over Cable**:

   - One end uses **T568A**, and the other end uses **T568B**.
   - Used to connect **similar devices** (e.g., PC ↔ PC, Switch ↔ Switch).

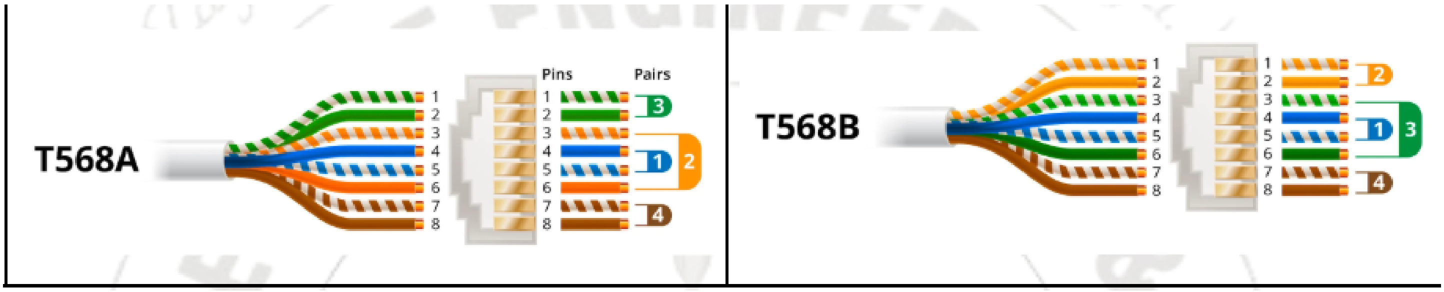
7. The **crimping tool** is used to press and secure the wires into the RJ-45 connector pins, ensuring proper contact.

8. After crimping, the cable is tested using a **LAN cable tester** to check continuity and correct pin configuration.

9. Proper crimping ensures **reliable data transfer**, reduced interference, and minimal signal loss.

Pin Configuration:

Pin	T568A Color	T568B Color
1	White/Green	White/Orange
2	Green	Orange
3	White/Orange	White/Green
4	Blue	Blue
5	White/Blue	White/Blue
6	Orange	Green
7	White/Brown	White/Brown
8	Brown	Brown



Apparatus / Components:

- Cat5e or Cat6 UTP Cable
- RJ-45 Connectors
- Crimping Tool
- Cable Stripper
- LAN Cable Tester

Procedure:

1. Strip about 1 inch of the UTP cable sheath.
2. Untwist and straighten the wire pairs.
3. Arrange wires as per **T568A** or **T568B** color code.
4. Cut all wires evenly and insert into the RJ-45 connector.
5. Crimp the connector firmly using the **crimping tool**.
6. Repeat the process for the other end of the cable.
7. Test using a **LAN cable tester** to verify connectivity.

Ethernet Crimping in Cisco Packet Tracer

Objective:

To understand and simulate the use of **Straight-through** and **Crossover** Ethernet cables between different network devices.

What You Can Do in CPT

1. Simulate Straight-Through Cable Connections

Purpose: Connect *different devices* (like PC ↔ Switch, Switch ↔ Router).

Steps:

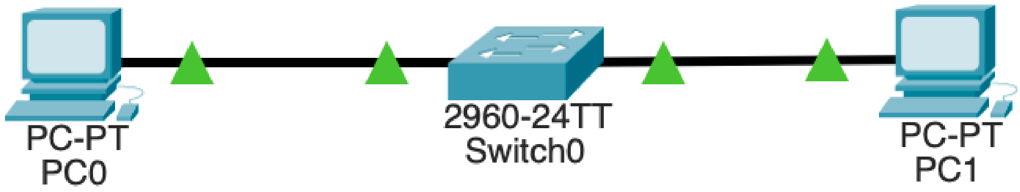
1. Open Cisco Packet Tracer.

2. Drag and drop:

   - 2 PCs (`End Devices`)
   - 1 Switch (`2960-24TT`)

3. Connect them using:

   ```
   Connections → Copper Straight-Through cable
   ```



2. Simulate Crossover Cable Connections

Purpose: Connect *similar devices* (like PC ↔ PC, Switch ↔ Switch). **Steps:**

1. Use:

   ```
   Connections → Copper Cross-Over cable
   ```

2. Connect PC0 ↔ PC1 (or Switch0 ↔ Switch1).

3. Observe that both link lights turn green.

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3. Assign IP Addresses

For PCs connected via cable:

1. Click on each PC → **Desktop** → **IP Configuration**

2. Example setup:

   ```
   PC0: 192.168.1.1  / 255.255.255.0
   PC1: 192.168.1.2  / 255.255.255.0
   ```

3. Ping test:

   ```
   PC0 → Command Prompt → ping 192.168.1.2
   ```

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4. Experiment with Wrong Cable Type

Try connecting:

- PC ↔ Switch using *Cross-over cable*
- PC ↔ PC using *Straight-through cable*

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5. Document the Color Codes

Inside CPT, you can't see the wire colors — but you can:

- Write a note on the workspace:

  ```
  T568A → White/Green, Green, White/Orange, Blue, White/Blue, Orange,
  White/Brown, Brown
  T568B → White/Orange, Orange, White/Green, Blue, White/Blue, Green,
  White/Brown, Brown
  ```

- Label each cable accordingly for demonstration.

🧠 Concepts You Demonstrate in CPT

Concept	Demonstrated By
Straight vs Cross Cable	Connecting different/similar devices
Cable Functionality	Green link lights + Ping test
Logical Connectivity	IP addressing + Communication
Importance of Wiring Standards	Wrong cable type = connection fail

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Optional Add-ons for Report:

Aim: To simulate Ethernet cable crimping and verify connectivity using Cisco Packet Tracer. **Result:** The straight-through and crossover cables were successfully tested between suitable devices.